Tongji Calculus, 7th Edition · Chapter 2

Derivatives and Differentials

Exercise Workbook

100 classic-method adaptations and high-quality synthesis problems

All mathematics compiled by XeTeX / LaTeX · 4:3 study edition

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

19 / 100

Classic-method variants

63 / 100

Original synthesis / diagnosis

18 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

The Derivative Concept 4 min · derivative definition · increments · limit

Basic Single choice Foundation Open-text method adaptation

Q001 · Recognizing the Definition of a Derivative

Suppose f is defined in a neighborhood of x_0. Which limit, if it exists, equals $f'(x_0)$?

A. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$

B. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{x_0}$

C. $\lim_{x \to x_0} \frac{f(x)-f(x_0)}{x}$

D. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)+f(x_0)}{\Delta x}$

Workspace · add pages as needed

The Derivative Concept 4 min · instantaneous velocity · average velocity · derivative definition

Basic **Fill in the blank** **Foundation** **Open-text method adaptation**

Q002 · Expressing Instantaneous Velocity as a Limit

A particle moves along a line with position $s = s(t)$. Its instantaneous velocity at $t = 3$ can be written as $v(3) =$ ____.

Workspace · add pages as needed

The Derivative Concept

7 min · first-principles differentiation · nonzero base point · polynomial

Basic **Calculation** **Foundation** **Open-text method adaptation**

Q003 · Differentiate at a Nonzero Point from First Principles

Using only the definition of the derivative, find the derivative of $f(x) = x^2$ at $x = 2$.

Workspace · add pages as needed

The Derivative Concept 5 min · differentiability and continuity · necessary condition · logical reasoning

Basic **True/false with justification** **Foundation** **Open-text method adaptation**

Q004 · Does Differentiability Guarantee Continuity?

Decide whether the statement is true and justify: if f is differentiable at x_0, then f is continuous at x_0.

Workspace · add pages as needed

The Derivative Concept 4 min · geometric meaning · tangent slope · point-slope form

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q005 · Writing a Tangent Line from a Derivative

Suppose $f(2) = 5$ and $f'(2) = -3$. Which equation is the tangent line to $y = f(x)$ at $(2, 5)$?

A. $y - 5 = -3(x - 2)$

B. $y - 2 = -3(x - 5)$

C. $y - 5 = (\frac{1}{3})(x - 2)$

D. $y + 5 = -3(x + 2)$

Workspace · add pages as needed

The Derivative Concept 6 min · one-sided derivatives · absolute-value function · differentiability test

Basic **Fill in the blank** **Foundation** **Open-text method adaptation**

Q006 · One-Sided Derivatives of an Absolute-Value Function

Let $f(x) = |x|$. Then $f_-'(0) =$ ____, $f_+'(0) =$ ____, so f is ____ at 0 (write "differentiable" or "not differentiable").

Workspace · add pages as needed

The Derivative Concept 8 min · first-principles differentiation · tangent line · normal line

Basic **Calculation** **Foundation** **Open-text method adaptation**

Q007 · Tangent and Normal Lines from First Principles

Let $P(1, 1)$ lie on the curve $y = x^2$. Use the definition of the derivative to find the tangent and normal lines at P.

Workspace · add pages as needed

Differentiation Rules 4 min · basic derivative formulas · logarithmic function · domain

Basic Single choice Foundation Open-text method adaptation

Q008 · Choose the Derivative of a Logarithm

For $x > 0$, what is the derivative of $y = \ln x$?

A. $y' = \frac{1}{x}$

B. $y' = \ln x$

C. $y' = x$

D. $y' = e^x$

Workspace · add pages as needed

Differentiation Rules 4 min · product rule · exponential function · single choice

Basic **Single choice** **Foundation** **Classic-method variant**

Q009 · A Polynomial Times an Exponential

Let $f(x) = x^2e^x$. Which expression equals $f'(x)$?

A. $2xe^x$

B. $e^x(x^2 + 2x)$

C. x^2e^x

D. $e^x(x + 2)$

Workspace · add pages as needed

Differentiation Rules

5 min · quotient rule · formula diagnosis · true or false

Basic **True/false with justification** **Foundation** **Original synthesis / diagnosis**

Q010 · Can a Quotient Be Differentiated Termwise?

True or false, with justification: If u and v are differentiable and $v^{'}(x) \neq 0$, then $[\frac{u(x)}{v(x)}]' = \frac{u'(x)}{v'(x)}$.

Workspace · add pages as needed

Differentiation Rules 5 min · basic derivative formulas · logarithm · inverse trigonometric function

Basic **Fill in the blank** **Foundation** **Classic-method variant**

Q011 · A Linear Combination of Three Basic Functions

For $x > 0$, if $y = 3e^x - 2\ln x + 5\arctan x$, then $y' =$ ____.

Workspace · add pages as needed

Differentiation Rules 6 min · chain rule · power function · composite function

Basic **Calculation** **Foundation** **Open-text method adaptation**

Q012 · The Fifth Power of a Linear Function

Differentiate $y = (2x - 1)^5$ and explicitly identify the outer and inner functions.

Workspace · add pages as needed

Differentiation Rules

6 min · quotient rule · exponential function · trigonometric function

Basic **Calculation** **Foundation** **Classic-method variant**

Q013 · A Sine Divided by an Exponential

Differentiate $y = \frac{\sin x}{e^x}$ and write the result in the simplest product form.

Workspace · add pages as needed

Differentiation Rules 5 min · inverse-function derivative · corresponding points · single choice

Basic **Single choice** **Foundation** **Classic-method variant**

Q014 · The Derivative of an Inverse at Corresponding Points

Suppose f has a differentiable inverse $g = f^{-1}$ near 2, with $f(2) = 5$ and $f'(2) = -3$. What is $g'(5)$?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

Workspace · add pages as needed

Higher Derivatives

4 min · higher derivatives · polynomial · true or false

Basic **True/false with justification** **Foundation** **Classic-method variant**

Q015 · Higher Derivatives of a Polynomial

True or false, with justification: If $y = x^4 - 3x + 2$, then $y^{(5)}(x) \equiv 0$.

Workspace · add pages as needed

Higher Derivatives

5 min · higher trigonometric derivatives · four-step cycle · fill in the blank

Basic **Fill in the blank** **Foundation** **Classic-method variant**

Q016 · The 2026th Derivative of Cosine

If $y = \cos x$, then $y^{(2026)} =$ ____.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

5 min · implicit differentiation · product rule · single choice

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q017 · Recognizing a First Implicit Derivative

On the curve $x^2 + xy + y^2 = 7$, wherever y is a differentiable function of x, which expression equals y'?

A. $-\frac{2x+y}{x+2y}$

B. $-\frac{2x+y}{2x+y}$

C. $\frac{2x+y}{x+2y}$

D. $-\frac{x+2y}{2x+y}$

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 7 min · implicit differentiation · circle · tangent line

Basic **Calculation** **Foundation** **Open-text method adaptation**

Q018 · The Tangent to a Circle

At the point $(3, 4)$ on $x^2 + y^2 = 25$, find $\frac{dy}{dx}$ and the equation of the tangent line.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 6 min · implicit differentiation · derivative at a point · fill in the blank

Basic **Fill in the blank** **Foundation** **Classic-method variant**

Q019 · Evaluating an Implicit Derivative at a Point

For $x^2 + xy = 6$, $\frac{dy}{dx}$ at $(2, 1)$ is ____.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

8 min · implicit function · vertical tangent · true or false

Basic **True/false with justification** **Foundation** **Classic-method variant**

Q020 · The Domain of an Implicit-Derivative Formula

True or false, with justification: From $x^2 + y^2 = 1$ we obtain $y' = -\frac{x}{y}$, so this formula gives a finite tangent slope at every point of the circle.

Workspace · add pages as needed

Differentials 5 min · definition of differential · linear principal part · single choice

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q021 · Recognizing the Differential of a Function

If $y = f(x)$ is differentiable at x and $dx = \Delta x$, which expression is the differential dy?

A. $dy = f'(x)dx$

B. $dy = \Delta y$ for every finite Δx

C. $dy = f(x)dx$

D. dy is independent of dx

Workspace · add pages as needed

Differentials 5 min · differential calculation · power function · fill in the blank

Basic Fill in the blank Foundation Classic-method variant

Q022 · Numerical Differential of a Power Function

If $y = x^3$, then at $x = 2$ with $dx = 0.01$, $dy =$ ____.

Workspace · add pages as needed

The Derivative Concept 8 min · first-principles differentiation · nonzero base point · rational function

Standard **Calculation** **Methods** **Open-text method adaptation**

Q023 · Derivative of the Reciprocal Function at Any Nonzero Point

Let $f(x) = \frac{1}{x}$. For any $a \neq 0$, use only the definition of the derivative to find $f'(a)$.

Workspace · add pages as needed

The Derivative Concept 7 min · one-sided derivatives · continuity · necessary and sufficient conditions

Standard **Multiple choice** **Methods** **Original synthesis / diagnosis**

Q024 · Logical Relations Involving Differentiability

Let x_0 be an interior point of the domain of f. Which statements are correct?

A. If $f'(x_0)$ exists, then $f'_-(x_0)$ and $f'_+(x_0)$ both exist and are equal

B. If f is continuous at x_0, then $f'(x_0)$ must exist

C. If $f'(x_0)$ exists, then f is continuous at x_0

D. If $f'_-(x_0)$ and $f'_+(x_0)$ are both finite, then $f'(x_0)$ must exist

Workspace · add pages as needed

The Derivative Concept 10 min · differentiability implies continuity · increment representation · rigorous proof

Standard Proof Methods Open-text method adaptation

Q025 · Prove Differentiability Implies Continuity via an Infinitesimal Remainder

Suppose f is differentiable at x_0. Prove that some $\alpha(\Delta x) \to 0$ satisfies $f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + \alpha(\Delta x)\Delta x$, and use it to prove continuity at x_0.

Workspace · add pages as needed

The Derivative Concept

10 min · piecewise function · parameter determination · differentiability conditions

Standard **Calculation** **Methods** **Classic-method variant**

Q026 · Matching Parameters at a Junction

Let $f(x) = \begin{cases} x^2, & x \le 1, \\ ax+b, & x > 1. \end{cases}$ Find a, b so that f is differentiable at $x = 1$, and find $f'(1)$.

Workspace · add pages as needed

The Derivative Concept 8 min · instantaneous velocity · first-principles differentiation · units and direction

Standard Fill in the blank Methods Classic-method variant

Q027 · Instantaneous Velocity at a Specified Time

A particle has position $s(t) = t^3 - 6t^2 + 9t$ meters, with t in seconds. Use a difference-quotient limit to find the instantaneous velocity at $t = 2$, namely $v(2) =$ ___.

Workspace · add pages as needed

The Derivative Concept 10 min · square-root function · first-principles differentiation · tangent and normal

Standard **Calculation** **Methods** **Classic-method variant**

Q028 · Tangent and Normal to a Square-Root Curve

Without using a ready-made derivative formula, find the derivative of $y = \sqrt{x}$ at $(4, 2)$ from the definition, and write the tangent and normal lines.

Workspace · add pages as needed

The Derivative Concept

6 min · continuity need not imply differentiability · counterexample · converse statement

Standard | True/false with justification | Methods | Classic-method variant

Q029 · Testing the Converse of Differentiability Implies Continuity

Decide whether the statement is true and justify rigorously: if f is continuous at x_0, then f is differentiable at x_0.

Workspace · add pages as needed

The Derivative Concept 8 min · translated absolute value · one-sided derivatives · corner

Standard **Proof** **Methods** **Open-text method adaptation**

Q030 · Prove a Translated Absolute Value Is Not Differentiable at Its Corner

Let a be any real number. Prove that $f(x) = |x - a|$ is continuous at $x = a$, but has left derivative -1 and right derivative 1 there, and hence is not differentiable.

Workspace · add pages as needed

Differentiation Rules

7 min · product rule · polynomial · algebraic simplification

Standard **Calculation** **Methods** **Classic-method variant**

Q031 · Product Rule and Simplification

Differentiate $y = (x^2 + 1)(x^3 - 2x)$, and simplify the result to a polynomial.

Workspace · add pages as needed

Differentiation Rules 6 min · chain rule · trigonometric function · composite function

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q032 · Identify the Inner and Outer Functions

If $y = \sin(3x^2 - 1)$, then $y' =$ ____.

Workspace · add pages as needed

Differentiation Rules

8 min · differentiation rules · inverse function · chain rule

Standard **Multiple choice** **Methods** **Original synthesis / diagnosis**

Q033 · Distinguish Four Differentiation Formulas

Assume all differentiability, domain, nonzero-denominator, and inverse-continuity conditions required by the formulas hold. For the inverse formula, let $y_0 = f(x_0)$ and $f^{'}(x_0) \neq 0$. Which formulas are correct?

A. $(fg)' = f'g + fg'$

B. $(\frac{f}{g})' = \frac{f'g - fg'}{g^2}$ for $(g \neq 0)$

C. $(f \circ g)'(x) = f'(x)g'(x)$

D. $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$

Workspace · add pages as needed

Differentiation Rules

7 min · quotient rule · domain · simplification

Standard **Calculation** **Methods** **Classic-method variant**

Q034 · Quotient Rule for a Rational Function

Differentiate $y = \frac{x^2+1}{x-1}$, and state where the derivative formula applies.

Workspace · add pages as needed

Differentiation Rules

9 min · product rule · quotient rule · logarithm

Standard **Calculation** **Methods** **Classic-method variant**

Q035 · A Product Nested Inside a Quotient

For $x > 0$ and $x \neq 1$, differentiate $y = \frac{(x^2+1)\ln x}{x-1}$, and give a result over a common denominator.

Workspace · add pages as needed

Differentiation Rules

6 min · arcsine · radical composite · single choice

Standard **Single choice** **Methods** **Classic-method variant**

Q036 · A Composite of Arcsine and a Square Root

For $0 < x < 1$, which expression is the derivative of $y = \arcsin\sqrt{x}$?

A. $\frac{1}{\sqrt{1-x}}$

B. $\frac{1}{2\sqrt{x}\sqrt{1-x}}$

C. $\frac{1}{2\sqrt{x}(1-x)}$

D. $-\frac{1}{2\sqrt{x}\sqrt{1-x}}$

Workspace · add pages as needed

Differentiation Rules 7 min · differentiation rules · condition checking · multiple choice

Standard Multiple choice Methods Original synthesis / diagnosis

Q037 · Distinguishing Four Differentiation Rules

Assume the functions involved are differentiable at the relevant points and all expressions are defined. Which statements are correct?

A. $[u(v(x))]' = u'(v(x))v'(x)$

B. $[u(x)v(x)]' = u'(x)v'(x)$

C. $[\frac{u(x)}{v(x)}]' = \frac{u'(x)v(x)-u(x)v'(x)}{v^2(x)}$, where $v(x) \neq 0$

D. If f has a differentiable inverse and $f'(f^{-1}(y)) \neq 0$, then $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Workspace · add pages as needed

Differentiation Rules

8 min · multilevel chain rule · logarithm · trigonometric composite

Standard **Calculation** **Methods** **Classic-method variant**

Q038 · A Trigonometric Square Inside a Logarithm

Differentiate $y = \ln[1 + \sin^2(3x)]$, listing the layers from outside to inside.

Workspace · add pages as needed

Differentiation Rules 9 min · logarithmic differentiation · variable power · composite function

Standard **Calculation** **Methods** **Classic-method variant**

Q039 · Both Base and Exponent Depend on x

Differentiate $y = (x^2 + 1)^{\sin x}$ using logarithmic differentiation, and state its domain.

Workspace · add pages as needed

Differentiation Rules 13 min · inverse derivative · derivative definition · proof

Standard **Proof** **Methods** **Classic-method variant**

Q040 · Proving the Inverse Derivative Formula from the Definition

Suppose f is differentiable at x_0 with $f^{'}(x_0) \neq 0$ and is one-to-one near x_0. Its inverse g is continuous at $y_0 = f(x_0)$. Prove from the derivative definition that $g^{'}(y_0) = \frac{1}{f'(x_0)}$.

Workspace · add pages as needed

Differentiation Rules

8 min · piecewise function · parameter differentiability · fill in the blank

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q041 · Determining Two Parameters in a Piecewise Function

Let $f(x) = \begin{cases} ax + b, & x < 1, \\ x^2 + 1, & x \geq 1. \end{cases}$ For f to be differentiable at $x = 1$, $a =$ ___ and $b =$ ___.

Workspace · add pages as needed

Higher Derivatives

9 min · second derivative · third derivative · exponential-trigonometric product

Standard **Calculation** **Methods** **Classic-method variant**

Q042 · Second and Third Derivatives of an Exponential-Cosine Product

Let $y = e^{2x} \cos x$. Find y'' and y'''.

Workspace · add pages as needed

Higher Derivatives 7 min · higher derivative of a product · counterexample · true or false

Standard **True/false with justification** **Methods** **Original synthesis / diagnosis**

Q043 · Does the nth Derivative of a Product Have Only Two Terms?

True or false, with justification: For every $n \geq 2$, if u and v have the needed derivatives, then $(uv)^{(n)} = u^{(n)}v + uv^{(n)}$.

Workspace · add pages as needed

Higher Derivatives 7 min · second derivative · third derivative · sum of functions

Standard **Calculation** **Methods** **Classic-method variant**

Q044 · Higher Derivatives of an Exponential Plus a Sine

Let $y = \sin(2x) + e^{3x}$. Find y'' and y'''.

Workspace · add pages as needed

Higher Derivatives 13 min · higher rational derivatives · mathematical induction · proof

Standard Proof Methods Classic-method variant

Q045 · The nth Derivative of the Reciprocal of a Linear Factor

For every integer $n \geq 0$, prove by induction that if $y = \frac{1}{x-a}, x \neq a$, then $y^{(n)} = \frac{(-1)^n n!}{(x-a)^{n+1}}$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 11 min · implicit differentiation · exponential composite · slope at a point

Standard **Calculation** **Methods** **Classic-method variant**

Q046 · First Derivative of an Exponential Implicit Curve

The curve $e^{xy} + x + y = 1$ passes through $(0, 0)$. Find the general expression for y' and the tangent slope at that point.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 11 min · implicit differentiation · trigonometric composite · tangent

Standard **Calculation** **Methods** **Original synthesis / diagnosis**

Q047 · Differentiating a Trigonometric Implicit Curve

The curve $\sin(x + y) = xy$ passes through $(0, 0)$. Find the general expression for y' and the tangent line at that point.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

9 min · parametric differentiation · tangent line · parameter value

Standard **Calculation** **Methods** **Classic-method variant**

Q048 · First Derivative and Tangent of a Parametric Curve

For $x = t^2 + 1$ and $y = t^3 - t$, find $\frac{dy}{dx}$ at $t = 1$ and write the tangent line.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

6 min · parametric equations · derivative formula · fill in the blank

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q049 · The Parametric Derivative Formula and Its Value

If $x = t^2$ and $y = t^4 + t$, then when $\frac{dx}{dt} \neq 0$, $\frac{dy}{dx} =$ ____; at $t = 1$ its value is ____.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

6 min · parametric curve · vertical tangent · condition recognition

Standard | Single choice | Methods | Original synthesis / diagnosis

Q050 · Condition for a Vertical Tangent of a Parametric Curve

Suppose the parameter derivatives exist at $t = t_0$. Which condition most directly indicates a vertical tangent?

A. $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$

B. $\frac{dx}{dt} \neq 0$ and $\frac{dy}{dt} = 0$

C. $\frac{dx}{dt} = \frac{dy}{dt} = 1$

D. $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, with no further analysis needed

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

9 min · related rates · sphere volume · units

Standard **Calculation** **Methods** **Open-text method adaptation**

Q051 · Volume Rate of an Expanding Sphere

The radius of a spherical balloon increases at $2\,\mathrm{cm\,s^{-1}}$. How fast is its volume increasing when the radius is $5\,\mathrm{cm}$?

Workspace · add pages as needed

Differentials

9 min · differentiability and derivative · equivalent definitions · true or false

Standard **True/false with justification** **Methods** **Classic-method variant**

Q052 · Differentiability and Existence of a Derivative in One Variable

True or false, with justification: For a one-variable function $y = f(x)$, differentiability at a point is equivalent to existence of the derivative there, and the differential coefficient is $f'(x)$.

Workspace · add pages as needed

Differentials

8 min · differential rules · logarithmic composite · numerical differential

Standard **Calculation** **Methods** **Classic-method variant**

Q053 · Differential of a Logarithmic Composite

Let $y = \ln(1 + x^2)$. Find dy, and evaluate it at $x = 1$ with $dx = 0.02$.

Workspace · add pages as needed

Differentials

8 min · linear approximation · square root · approximation

Standard Calculation Methods Open-text method adaptation

Q054 · Approximating a Square Root with Differentials

Using only differential linearization near $x_0 = 4$, approximate $\sqrt{4.04}$.

Workspace · add pages as needed

Differentials 9 min · increment and differential · relative differential · multiple choice

Standard **Multiple choice** **Methods** **Original synthesis / diagnosis**

Q055 · Correct Statements about Differential Estimates

Let $y = f(x)$ be differentiable at x with $y \neq 0$. Which statements are correct? Select all that apply.

A. As $\Delta x \to 0, \Delta y = dy + o(\Delta x)$

B. dy is linear in dx

C. $\Delta y = dy$ for every finite dx

D. $\frac{dy}{y}$ is a first-order estimate of the relative increment $\frac{\Delta y}{y}$

Workspace · add pages as needed

The Derivative Concept 12 min · oscillatory function · first-principles differentiation · piecewise derivative

Advanced **Calculation** **Methods** **Original synthesis / diagnosis**

Q056 · Differentiability of an Oscillatory Function at the Origin

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ Decide whether f is differentiable at 0, and find the complete derivative $f'(x)$.

Workspace · add pages as needed

The Derivative Concept 10 min · even function · derivative at the origin · symmetric difference quotient

Advanced **Proof** **Methods** **Classic-method variant**

Q057 · Slope of an Even Function at the Origin

Suppose f is even and differentiable at $x = 0$. Prove that $f'(0) = 0$.

Workspace · add pages as needed

The Derivative Concept

8 min · modified difference quotient · derivative definition · linear combination

Advanced | Single choice | Methods | Original synthesis / diagnosis

Q058 · Recognizing the Limit of an Asymmetric Secant

Suppose f is differentiable at x_0. Which limit must equal $f'(x_0)$?

A. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0-h)}{3h}$

B. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0-h)}{2h}$

C. $\lim_{h\to 0} \frac{f(x_0+h)-f(x_0-h)}{h}$

D. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0)}{h}$

Workspace · add pages as needed

The Derivative Concept 12 min · parameter classification · absolute value · local expression

Advanced **Calculation** **Methods** **Original synthesis / diagnosis**

Q059 · A Moving Corner and Parameter Classification

Let $f_{a,b}(x) = |x - a| + bx$, where $a, b \in \mathbb{R}$. Determine all pairs (a,b) for which $f_{a,b}$ is differentiable at $x = 0$, and find $f'_{a,b}(0)$.

Workspace · add pages as needed

Differentiation Rules 8 min · inverse-function differentiation · corresponding points · reciprocal relation

Advanced Calculation Methods Classic-method variant

Q060 · Derivative of an Inverse at a Corresponding Point

Let $f(x) = x^3 + x$ and $g = f^{-1}$. Given that g is differentiable at $y = 2$, find $g'(2)$.

Workspace · add pages as needed

Differentiation Rules

14 min · proof of the product rule · derivative definition · adding and subtracting an intermediate term

Advanced **Proof** **Methods** **Classic-method variant**

Q061 · Prove the Product Rule from First Principles

Suppose u and v are differentiable at x_0. Using only the definition of the derivative, prove $(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0)$.

Workspace · add pages as needed

Differentiation Rules

12 min · multilayer chain rule · quotient rule · exponential and trigonometric functions

Advanced **Calculation** **Methods** **Classic-method variant**

Q062 · Combined Multilayer Chain and Quotient Rules

Differentiate $y = \frac{e^{\sin(x^2)}}{1+x^3}$, and state its domain.

Workspace · add pages as needed

The Derivative Concept

14 min · motion model · average velocity · instantaneous velocity · tangent interpretation

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q063 · From Average Velocity to Instantaneous Direction

A particle has position $s(t) = t^3 - 3t^2 + 2t$ meters, with t in seconds. (1) Find the average velocity on $[1, 1+h]$ for $h \neq 0$. (2) Use the definition to find $v(1)$. (3) Write the tangent to the position-time curve at $(1, s(1))$. (4) Interpret the sign of $v(1)$.

Workspace · add pages as needed

The Derivative Concept 10 min · endpoint derivative · unbounded difference quotient · error diagnosis

Advanced **Error diagnosis** **Synthesis** **Original synthesis / diagnosis**

Q064 · The Error in an "Infinite Derivative"

The function $f(x)=\sqrt{x}$ has domain $[0,+\infty)$. A student writes: "$f'(0)=\lim_{h\to 0}\frac{\sqrt{h}-0}{h}=\lim_{h\to 0}\frac{1}{\sqrt{h}}=+\infty$, so $f'(0)=+\infty$." Identify every key error and give the correct conclusion.

Workspace · add pages as needed

Differentiation Rules 8 min · basic formulas · domains · multiple choice

Advanced **Multiple choice** **Synthesis** **Original synthesis / diagnosis**

Q065 · Checking Both Formula and Domain

Which differentiation formulas are correct on the indicated domains?

A. $(\ln|x|)' = \frac{1}{x}, x \neq 0$

B. $(\arccos x)' = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

C. $(a^x)' = a^x \ln a, a > 0$ and $a \neq 1$

D. $[\sqrt{1-x^2}]' = -\frac{x}{\sqrt{1-x^2}}, -1 < x < 1$

Workspace · add pages as needed

Differentiation Rules

11 min · multilevel composite · arctangent · exponential-trigonometric composite

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q066 · Chain Rule for a Four-Layer Composite

Differentiate $y = \arctan\left(e^{\sin(x^2)}\right)$, using layer variables to explain the source of every factor.

Workspace · add pages as needed

Differentiation Rules 12 min · real-power derivative · logarithmic differentiation · proof

Advanced **Proof** **Synthesis** **Classic-method variant**

Q067 · Proving the Derivative Rule for Real Powers

Let α be any real number. Using $x^\alpha = e^{\alpha \ln x}$, prove that $(x^\alpha)' = \alpha x^{\alpha-1}$ for $x > 0$.

Workspace · add pages as needed

Differentiation Rules

15 min · piecewise function · parameter classification · continuity and differentiability

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q068 · Differentiability Classification for a Three-Parameter Piecewise Function

Let $f(x) = \begin{cases} x^2 + ax + b, & x < 1, \\ c \ln x + 2, & x \geq 1. \end{cases}$ Find all real triples (a, b, c) for which f is differentiable at $x = 1$, and give $f'(1)$.

Workspace · add pages as needed

Differentiation Rules

12 min · derivative of arctangent · inverse function · proof

Advanced **Proof** **Synthesis** **Classic-method variant**

Q069 · Deriving the Derivative of Arctangent from the Inverse Formula

View arctan x as the inverse of tan y on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and prove that $(\arctan x)' = \frac{1}{1+x^2}$.

Workspace · add pages as needed

Higher Derivatives

10 min · higher logarithmic derivatives · pattern recognition · nth derivative

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q070 · The nth Derivative of the Natural Logarithm

For $x > 0$, compute the first four derivatives of ln x and then state its nth derivative for $n \geq 1$.

Workspace · add pages as needed

Higher Derivatives 18 min · higher exponential-trigonometric derivatives · phase representation · mathematical induction

Advanced **Proof** **Synthesis** **Classic-method variant**

Q071 · A Unified nth-Derivative Formula for an Exponential-Cosine Product

Let $a^2+b^2>0, r=\sqrt{a^2+b^2}$, and choose θ so that $\cos\theta=\frac{a}{r}$ and $\sin\theta=\frac{b}{r}$. Prove by induction that $y=e^{ax}\cos(bx+\varphi)$ satisfies $y^{(n)}=r^n e^{ax}\cos(bx+\varphi+n\theta)$.

Workspace · add pages as needed

Higher Derivatives 16 min · higher derivative of a product · trigonometric cycle · comprehensive

Advanced | Parameter / synthesis / application | Synthesis | Classic-method variant

Q072 · The General Derivative of $x^2 \sin x$

Let $y = x^2 \sin x$. For $n \geq 2$, find an explicit formula for $y^{(n)}$ and check it at $n = 2$.

Workspace · add pages as needed

Higher Derivatives 13 min · higher rational derivatives · partial fractions · nth derivative

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q073 · Decompose Before Taking Higher Derivatives of a Rational Function

Let $y = \frac{1}{(x-1)(x+2)}$, $x \neq 1, -2$. Find $y^{(n)}$ for $n \geq 0$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

18 min · second implicit derivative · quadratic curve · successive differentiation

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q074 · Second Derivative of a Quadratic Implicit Curve

The curve $x^2 + xy + y^2 = 7$ passes through $(1, 2)$. Find y' and y'' at that point.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

14 min · second parametric derivative · chain-rule differentiation · point value

Advanced Calculation Synthesis Classic-method variant

Q075 · Second Derivative of a Parametric Curve

For $x = t^2 + 1$ and $y = t^3$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $t = 1$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 9 min · related rates · units · sign · multiple choice

Advanced | Multiple choice | Synthesis | Original synthesis / diagnosis

Q076 · Modeling Principles for Related Rates

Which statements about related-rate problems are correct? Select all that apply.

A. First write a geometric or physical relation valid over time, then differentiate with respect to t

B. If a length is decreasing, its rate is negative under the chosen positive direction

C. The unit of $\frac{dV}{dt}$ is $\frac{\text{length}^3}{\text{time}}$

D. Current numerical values may be substituted before differentiation and treated as constants because only one instant matters

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 18 min · related rates · similar cones · volume · units

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q077 · Radius and Volume Rates in a Conical Water Surface

Water in an inverted conical tank forms similar cones satisfying $h = 3r$, with h, r measured in cm. The depth increases at $\frac{dh}{dt} = 6\,\text{cm}\,\text{min}^{-1}$. When $h = 6\,\text{cm}$, find $\frac{dr}{dt}$ and $\frac{dV}{dt}$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 20 min · related rates · ladder model · angular velocity · units

Advanced Parameter / synthesis / application Synthesis Open-text method adaptation

Q078 · Linear and Angular Rates of a Sliding Ladder

A $10\,\mathrm{m}$ ladder leans against a wall. Its foot moves away from the wall at $\frac{dx}{dt} = 0.6\,\mathrm{m\,s^{-1}}$. Let x be the foot's distance from the wall, y the top height, and θ the angle with the ground. When $x = 6\,\mathrm{m}$, find $\frac{dy}{dt}$ and $\frac{d\theta}{dt}$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

17 min · general implicit formula · chain rule · proof

Advanced **Proof** **Synthesis** **Classic-method variant**

Q079 · A General Derivative Formula for a Separable Implicit Equation

Let φ and ψ be differentiable, and suppose $y = y(x)$ satisfies $\varphi(x) + \psi(y) = C$ with $\psi'(y) \neq 0$. Prove $y' = -\frac{\varphi'(x)}{\psi'(y)}$, and explain the role of $\psi'(y) \neq 0$.

Workspace · add pages as needed

Differentials 13 min · absolute differential · relative differential · measurement error · units

Advanced **Calculation** **Synthesis** **Open-text method adaptation**

Q080 · Absolute and Relative Error Estimates for a Circle's Area

A circle's radius is measured as $r = 10$ cm with possible absolute measurement error $|dr| \leq 0.02$ cm. Use differentials to estimate the maximum absolute and relative errors in $A = \pi r^2$.

Workspace · add pages as needed

Differentials

11 min · linear approximation · negative power · approximation limits

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q081 · Approximating a Reciprocal Square Root

Use differential linearization at $x_0 = 1$ to approximate $\frac{1}{\sqrt{0.98}}$, and state whether this method provides a rigorous error bound.

Workspace · add pages as needed

Differentials 18 min · linear approximation · relative differential · approximation comparison · synthesis

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q082 · Linear Prediction and Actual Discrepancy for a Fifth Power

Let $y = x^5$. Using $x_0 = 1$ and $dx = 0.02$, approximate $(1.02)^5$ by differentials and give the predicted relative increment. Then use the exact value 1.1040808032 from direct multiplication to find exact minus approximate, and explain why this is not an error bound supplied by this chapter.

Workspace · add pages as needed

The Derivative Concept 12 min · parameterized power function · first-principles differentiation · one-sided limits

Hard True/false with justification Synthesis Classic-method variant

Q083 · Complete Classification of a Parameterized Power Corner

Decide whether the following statement is true and prove it: for $\alpha > 0$, $f_\alpha(x) = |x|^\alpha$ is differentiable at $x = 0$ if and only if $\alpha > 1$; when differentiable, $f_\alpha'(0) = 0$.

Workspace · add pages as needed

The Derivative Concept 18 min · oscillatory function · existence of a tangent · discontinuous derivative · sequence test

Hard **Parameter / synthesis / application** **Synthesis** **Original synthesis / diagnosis**

Q084 · Stable Secants but Uncontrolled Nearby Tangents

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x^2}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ (1) Prove continuity at 0. (2) Prove differentiability at 0 and write the tangent there. (3) Find $f'(x)$ for $x \neq 0$. (4) Construct $x_n \to 0^+$ showing that f' is not continuous at 0.

Workspace · add pages as needed

Differentiation Rules 14 min · inverse trigonometric function · exponential function · quotient rule · chain rule

Hard **Calculation** **Synthesis** **Classic-method variant**

Q085 · Structural Simplification after Nested Rules

Differentiate $y = \arctan[\frac{e^x-1}{e^x+1}]$, and simplify the result so that it contains no compound fraction.

Workspace · add pages as needed

Differentiation Rules 16 min · inverse-function derivative theorem · reciprocal difference quotient · corresponding increments

Hard **Proof** **Synthesis** **Classic-method variant**

Q086 · Prove the Inverse-Function Derivative Formula from First Principles

Suppose f is one-to-one on a neighborhood U of x_0, its image contains a neighborhood of $y_0 = f(x_0)$, and it has inverse $x = g(y)$ on that image. Assume f is differentiable at x_0, $f^{'}(x_0) \neq 0$, and $g(y) \to x_0$ as $y \to y_0$. Prove $g^{'}(y_0) = \frac{1}{f'(x_0)}$.

Workspace · add pages as needed

Higher Derivatives

16 min · higher product derivative · polynomial-exponential product · finite sum

Hard **Calculation** **Synthesis** **Classic-method variant**

Q087 · The nth Derivative of a Cubic Times an Exponential

Let $y = x^3 e^{2x}$. For $n \geq 3$, find $y^{(n)}$ with no summation sign in the answer.

Workspace · add pages as needed

Higher Derivatives

20 min · Leibniz formula · mathematical induction · binomial coefficients

Hard **Proof** **Synthesis** **Classic-method variant**

Q088 · Inductive Proof of the Higher Product Formula

Suppose u and v have the required derivatives. Prove by induction that for $n \geq 0, (uv)^{(n)} = \sum_{k=0}^{n} C(n,k)u^{(k)}v^{(n-k)}$.

Workspace · add pages as needed

Higher Derivatives 14 min · error diagnosis · higher logarithmic derivatives · chain factors

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q089 · Diagnosing a Missing Repeated Chain Factor

For $y = \ln(2x-1), x > \frac{1}{2}, a$ student writes $y^{(n)} = \frac{(-1)^{n-1}(n-1)!}{(2x-1)^n}$. Identify the error, give the correct formula, and check at least $n = 1$ and $n = 2$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

24 min · second implicit derivative · cubic curve · hard calculation

Hard **Calculation** **Synthesis** **Classic-method variant**

Q090 · Second Derivative of a Cubic Implicit Curve

The curve $x^3 + y^3 = 6xy$ passes through $(3, 3)$. Find y' and y'' at that point.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

23 min · second parametric derivative formula · quotient rule · proof

Hard **Proof** **Synthesis** **Classic-method variant**

Q091 · Deriving the Second Parametric Derivative Formula

Let $x = x(t)$ and $y = y(t)$ be twice differentiable with $x^{'}(t) \neq 0$. Prove $\frac{d^2y}{dx^2} = \frac{x'(t)y''(t)-y'(t)x''(t)}{[x'(t)]^3}$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates

22 min · related-rate error · similar cones · unit diagnosis

Hard **Error diagnosis** **Synthesis** **Classic-method variant**

Q092 · Diagnosing a Frozen Water-Surface Radius

An inverted conical tank is $9\,\text{m}$ high with rim radius $3\,\text{m}$. The water depth rises at $0.2\,\text{m}\,\text{min}^{-1}$. Find the volume rate when $h = 3\,\text{m}$. A student fixes the water radius at $3\,\text{m}$, writes $V = \frac{1}{3}\pi \cdot 3^2 h$, and obtains $\frac{dV}{dt} = 0.6\pi\,\text{m}^3\,\text{min}^{-1}$. Diagnose the error and give the correct result.

Workspace · add pages as needed

Differentials 24 min · definition of differentiability · definition of derivative · if and only if · small-o proof

Hard **Proof** **Synthesis** **Classic-method variant**

Q093 · Equivalence between a Linear Increment Decomposition and a Derivative

Prove that a one-variable function f is differentiable at x_0 if and only if there is a constant A such that $\Delta y = A\Delta x + o(\Delta x)$. Prove that $A = f^{'}(x_0)$, so $dy = Adx$.

Workspace · add pages as needed

Differentials 20 min · linearization error · base-point derivative · approximation boundary

Hard Error diagnosis Synthesis Classic-method variant

Q094 · Diagnosing a Wrong Base Point and an Exact Equality

To approximate $\sqrt{25.5}$, a student takes $f(x) = \sqrt{x}, x_0 = 25$, and $\Delta x = 0.5$, but writes $dy = f'(25.5)\Delta x = \frac{0.5}{2\sqrt{25.5}} \approx 0.04951$ and claims $\sqrt{25.5} = 5.04951$. Identify all key errors, give the correct differential approximation, and state whether a rigorous error bound follows.

Workspace · add pages as needed

Higher Derivatives 17 min · very high derivative · phase cycle · comprehensive calculation

Hard Parameter / synthesis / application Challenge Classic-method variant

Q095 · Evaluating the 2026th Derivative at Zero

Let $y = e^x(\cos x + \sin x)$. Without expanding derivative by derivative, find $y^{(2026)}(0)$ and explain the cyclic structure used.

Workspace · add pages as needed

Higher Derivatives 17 min · repeated linear factor · partial fractions · higher rational derivatives

Hard Calculation Challenge Classic-method variant

Q096 · The nth Derivative of a Rational Function with a Repeated Pole

Let $y = \frac{1}{x^2(x-1)}, x \neq 0, 1$. Find $y^{(n)}$ for $n \geq 0$.

Workspace · add pages as needed

The Derivative Concept 28 min · two-parameter classification · oscillatory function · differentiability · continuity of the derivative

Challenge **Proof** **Challenge** **Original synthesis / diagnosis**

Q097 · Critical Exponents in a Two-Parameter Oscillatory Family

Let $\alpha, \beta > 0$ and $f_{\alpha,\beta}(x) = \begin{cases} |x|^\alpha \sin\left(\frac{1}{|x|^\beta}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ (1) Prove that $f_{\alpha,\beta}$ is continuous at 0. (2) Completely classify differentiability at 0. (3) When it is differentiable, completely classify continuity of $f'_{\alpha,\beta}$ at 0.

Workspace · add pages as needed

Higher Derivatives 24 min · squared logarithm · harmonic numbers · inductive proof

Challenge Proof Challenge Original synthesis / diagnosis

Q098 · The nth Derivative of a Squared Logarithm and Harmonic Numbers

Define $H_0 = 0$ and $H_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$ for $m \geq 1$. For $x > 0$, prove by induction that for $n \geq 1$, $[(\ln x)^2]^{(n)} = 2(-1)^{n-1}(n-1)!x^{-n}[\ln x - H_{n-1}]$.

Workspace · add pages as needed

Implicit and Parametric Derivatives; Related Rates 32 min · general second implicit derivative · higher chain rule · challenge proof

Challenge Proof Challenge Classic-method variant

Q099 · A General Second-Derivative Formula for a Separable Implicit Equation

Let φ and ψ be twice differentiable, with $y = y(x)$ satisfying $\varphi(x) + \psi(y) = C$ and $\psi'(y) \neq 0$. Prove $y'' = -\frac{\varphi''(x)}{\psi'(y)} - \frac{[\varphi'(x)]^2\psi''(y)}{[\psi'(y)]^3}$ using only the differentiation rules of this chapter.

Workspace · add pages as needed

Differentials 30 min · product rule for differentials · increment expansion · definition proof · challenge

Challenge **Proof** **Challenge** **Classic-method variant**

Q100 · Proving the Product Rule for Differentials from the Definition

Let $u = u(x)$ and $v = v(x)$ be differentiable at x_0. Using only increment decompositions, prove that uv is differentiable at x_0 and $d(uv) = v\,du + u\,dv$.

Do not cite the mean value theorem or later-chapter tools.

Workspace · add pages as needed